

Pioneer River Water, Salt and Total Nitrogen Budget Calculation Methodology

Whole Domain and Main Control Volume (Main CV) Closure Framework

This report defines the calculation procedure used to check Water, Salt and Total Nitrogen (TN) budget closure. The emphasis is on **calculation flow, source files, equations, variables, sign conventions and units**. No particular closure result is reproduced here.

The common question is:

Does the inventory change inside the control volume agree with the integrated RHS accounting?

For any budget quantity X ,

$$R_X = \Delta X_{\text{LHS}} - \Delta X_{\text{RHS}}$$

$$R_X(\%) = 100 \frac{R_X}{|\Delta X_{\text{RHS}}|}$$

1. Common calculation framework

1.1 Control volumes and calculation pathways

Two spatial scopes are used.

- **Whole Domain:** Main CV + river buffer + ocean buffer.
- **Main CV:** the nested regional control volume used for regional budget accounting.

The framework allows up to two independent LHS inventory pathways and two alternative RHS transport pathways, depending on the saved outputs and the calculation adopted for each budget quantity.

Whole Domain

- LHS-1: node-based inventory reconstruction.
- LHS-2: element-layer inventory reconstruction.
- RHS Method B: reconstructed flux across the exterior boundary edges.
- RHS Method C: not used in the present Whole-Domain framework.

Main CV

- LHS-1: node-based inventory reconstruction.
- LHS-2: element-layer inventory reconstruction.
- RHS Method B: reconstructed flux across the Main-CV interface edges.
- RHS Method C: native SCHISM region-interface transport from outputs/flux.out.

In the present methodology, Water and TN retain both LHS-1 and LHS-2 as independent inventory checks. Salt uses only the direct node-based LHS-1 inventory calculation; no additional element-layer Salt inventory pathway is introduced. Method B and Method C are alternative boundary-transport calculations for the Main CV and are not added together. The exact variables used by each pathway are defined in the corresponding budget section.

1.2 Region convention and transport sign

For the Main CV native flux.out accounting:

- Region 0 = river buffer.
- Region 1 = Main CV.
- Region 2 = ocean / outside.

All post-processing is converted to one common convention:

positive transport = transport INTO the selected control volume

A negative cumulative transport therefore represents net export.

For the Main CV native interface columns:

$$Q_{\text{River,in}} = -Q_{0/1,\text{raw}}$$

$$Q_{\text{Ocean,in}} = +Q_{1/2,\text{raw}}$$

The River column changes sign because native positive transport for the Region-0/1 interface is oriented from Region 1 to Region 0. The Ocean column retains its sign because native positive transport for the Region-1/2 interface is oriented from Region 2 to Region 1.

1.3 Common symbols and units

The notation used throughout the report is:

- e : horizontal element index.
- s : edge index.
- k : vertical layer / interval index.
- i, j : the two nodes forming edge s .
- A_e : horizontal area of element e [m^2].
- L_s : plan-view length of edge s [m].
- D_e : element dry flag; 0 = wet and 1 = dry.
- W_s : wet-edge mask.
- Q : water discharge / volume flux [$\text{m}^3 \text{s}^{-1}$].
- ΔV : water-volume change [m^3].
- S : salinity [PSU].
- M_S : equivalent salt mass [tonnes].
- C_{TN} : TN concentration [mmol N m^{-3}].
- M_{TN} : TN mass [t N].

1.4 Time integration and time units

Boundary fluxes are rates per second. For a sampled time series $F_n = F(t_n)$, cumulative transport is calculated with the trapezoidal rule:

$$I_F = \sum_{n=1}^{N-1} \frac{F_n + F_{n+1}}{2} \Delta t_n$$

When the time column is stored in **days** but the flux is per **second**, the time increment used in the integration must be converted to seconds:

$$\Delta t_n^{(s)} = 86400(t_{n+1}^{(d)} - t_n^{(d)})$$

Therefore:

- Water flux [$\text{m}^3 \text{s}^{-1}$] integrated with $\Delta t^{(s)}$ gives [m^3].
- Salt transport [$\text{PSU m}^3 \text{s}^{-1}$] integrated with $\Delta t^{(s)}$ gives [PSU m^3] before conversion to tonnes.
- TN boundary transport [mmol N s^{-1}] integrated with $\Delta t^{(s)}$ gives [mmol N] before conversion to t N.

AED process diagnostics with native units per day are integrated using Δt in **days**, not seconds.

2. Water budget

Water storage is a geometric quantity. The LHS is the change in stored water volume. The RHS is the cumulative net water volume transported across the selected boundary.

$$\Delta V_{\text{LHS}} = V(t_{\text{end}}) - V(t_{\text{begin}}) \quad [\text{m}^3]$$

$$\Delta V_{\text{RHS}} = \int_{t_{\text{begin}}}^{t_{\text{end}}} Q_{\text{net}}(t) dt \quad [\text{m}^3]$$

2.1 LHS-1: SCHISM node-based storage

LHS-1 reconstructs water volume from nodal bathymetry, nodal water-surface elevation and horizontal element geometry.

Source files

- out2d_*.nc: nodal elevation $\eta_i(t)$ [m] and dryFlagElement.
- hgrid.gr3: nodal bathymetric depth d_i [m], node coordinates and element connectivity.

Step 1 — Calculate water depth at each node

$$H_i(t) = d_i + \eta_i(t) \quad [\text{m}]$$

Step 2 — Calculate the mean water depth of each triangular element

For element e with nodes 1, 2 and 3:

$$\bar{H}_e(t) = \frac{H_1(t) + H_2(t) + H_3(t)}{3} \quad [\text{m}]$$

Step 3 — Calculate horizontal element area

For node coordinates (x_1, y_1) , (x_2, y_2) and (x_3, y_3) :

$$A_e = \frac{|(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)|}{2} \quad [\text{m}^2]$$

Step 4 — Apply the element wet/dry mask

dryFlagElement is represented by $D_e(t)$:

$$D_e(t) = 0 \quad \text{wet}, \quad D_e(t) = 1 \quad \text{dry}$$

The wet-element multiplier is:

$$1 - D_e(t)$$

Step 5 — Calculate water volume of each element

$$V_e(t) = [1 - D_e(t)] A_e \bar{H}_e(t) \quad [\text{m}^3]$$

Step 6 — Sum all elements in the selected control volume

$$V_{CV}(t) = \sum_{e=1}^{N_e} V_e(t) \quad [\text{m}^3]$$

Step 7 — Calculate the LHS storage change

$$\Delta V_{LHS,1} = V_{CV}(t_{\text{end}}) - V_{CV}(t_{\text{begin}}) \quad [\text{m}^3]$$

2.2 LHS-2: AED element-layer storage

LHS-2 calculates the same water storage independently from the AED element-layer thickness.

Source files

- aed_data_cmb_*.nc: ENV_layer_ht(t,e,k) [m].
- hgrid.gr3: element area A_e [m²].
- out2d_*.nc: dryFlagElement.

Step 1 — Read the thickness of each AED layer

$$h_{e,k}(t) = \text{ENV_layer_ht}(t, e, k) \quad [\text{m}]$$

Step 2 — Sum the active AED layers in one element

$$H_{e,AED}(t) = \sum_k h_{e,k}(t) \quad [\text{m}]$$

Step 3 — Calculate the water volume of one element

$$V_{e,AED}(t) = [1 - D_e(t)] A_e H_{e,AED}(t) \quad [\text{m}^3]$$

Step 4 — Sum all elements

$$V_{CV,AED}(t) = \sum_e V_{e,AED}(t) \quad [\text{m}^3]$$

Step 5 — Calculate the LHS storage change

$$\Delta V_{LHS,2} = V_{CV,AED}(t_{\text{end}}) - V_{CV,AED}(t_{\text{begin}}) \quad [\text{m}^3]$$

The storage consistency check is:

$$\Delta V_{LHS,1} \approx \Delta V_{LHS,2}$$

2.3 RHS Method B: node-based reconstruction of edge-layer water flux

Method B reconstructs water transport from saved nodal velocity, vertical coordinates and horizontal edge geometry. The same sequence is used for Whole-Domain exterior boundary edges and Main-CV interface edges.

Source files

- out2d_*.nc: dryFlagElement.

- horizontalVelX_*.nc: nodal u velocity [m s^{-1}].
- horizontalVelY_*.nc: nodal v velocity [m s^{-1}].
- zCoordinates_*.nc: nodal vertical coordinates z [m].
- hgrid.gr3: node coordinates, connectivity, edge length and edge-normal geometry.
- fluxflag.prop + hgrid.gr3: Main-CV region/interface edge identification.

Step 1 — Determine whether an edge is active and wet

For an internal edge s shared by elements e_L and e_R :

$$W_s(t) = [1 - D_{e_L}(t)][1 - D_{e_R}(t)]$$

Thus $W_s = 1$ means the edge is active and wet; $W_s = 0$ excludes the edge. For an exterior boundary edge with one adjacent element e :

$$W_s(t) = 1 - D_e(t)$$

Step 2 — Read nodal velocity at each vertical level

At node i and level k :

$$\begin{aligned} u_{i,k}(t) & \quad [\text{m s}^{-1}] \\ v_{i,k}(t) & \quad [\text{m s}^{-1}] \end{aligned}$$

The same quantities are read at node j .

Step 3 — Calculate edge length

For edge s joining nodes $i(x_i, y_i)$ and $j(x_j, y_j)$:

$$L_s = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2} \quad [\text{m}]$$

Step 4 — Define the unit normal to the edge

$$\begin{aligned} n_x &= \frac{y_j - y_i}{L_s} \\ n_y &= \frac{x_i - x_j}{L_s} \end{aligned}$$

The normal orientation is selected consistently with the positive-into-control-volume convention.

Step 5 — Project nodal velocity onto the edge-normal direction

At node i :

$$u_{n,i,k} = u_{i,k}n_x + v_{i,k}n_y \quad [\text{m s}^{-1}]$$

At node j :

$$u_{n,j,k} = u_{j,k}n_x + v_{j,k}n_y \quad [\text{m s}^{-1}]$$

Average the two nodal values to the edge:

$$u_{n,s,k} = \frac{u_{n,i,k} + u_{n,j,k}}{2} \quad [\text{m s}^{-1}]$$

Step 6 — Calculate layer thickness at the two edge nodes

$$\begin{aligned} h_{i,k} &= |z_{i,k+1} - z_{i,k}| \quad [\text{m}] \\ h_{j,k} &= |z_{j,k+1} - z_{j,k}| \quad [\text{m}] \end{aligned}$$

Average the two nodal thicknesses to the edge:

$$h_{s,k} = \frac{h_{i,k} + h_{j,k}}{2} \quad [\text{m}]$$

Step 7 — Calculate the cross-sectional area of one edge layer

$$A_{s,k}^{\text{section}} = L_s h_{s,k} \quad [\text{m}^2]$$

Step 8 — Calculate water flux through one edge layer

$$q_{s,k}(t) = W_s(t) u_{n,s,k}(t) A_{s,k}^{\text{section}}(t) \quad [\text{m}^3 \text{s}^{-1}]$$

Unit check:

$$[\text{m s}^{-1}] \times [\text{m}^2] = [\text{m}^3 \text{s}^{-1}]$$

Step 9 — Sum all active layers of one edge

$$Q_s(t) = \sum_k q_{s,k}(t) \quad [\text{m}^3 \text{s}^{-1}]$$

Step 10 — Sum all selected boundary edges

$$Q_{\text{net}}(t) = Q_1(t) + Q_2(t) + \dots + Q_{N_b}(t) \quad [\text{m}^3 \text{s}^{-1}]$$

Step 11 — Integrate over time

$$\Delta V_{\text{RHS,B}} = \int_{t_{\text{begin}}}^{t_{\text{end}}} Q_{\text{net}}(t) dt \quad [\text{m}^3]$$

2.4 RHS Method C: native SCHISM region-interface water flux

Method C is used for the Main CV. SCHISM has already accumulated region-interface transport during runtime, so no nodal velocity reconstruction is performed.

Source files

- fluxflag.prop: Region 0 / Region 1 / Region 2 element labels.
- outputs/flux.out: native region-interface transport time series.

For each iflux = 2 time block, the relevant fields are:

- Row 0: VOL net $[\text{m}^3 \text{s}^{-1}]$.
- Row 1: VOL positive $[\text{m}^3 \text{s}^{-1}]$.
- Row 2: VOL negative $[\text{m}^3 \text{s}^{-1}]$.
- Rows 3–4: temperature transport, positive / negative $[\text{deg C m}^3 \text{s}^{-1}]$.
- Rows 5–6: salinity transport, positive / negative $[\text{PSU m}^3 \text{s}^{-1}]$.
- Following pairs: AED tracer transport, positive / negative $[\text{native tracer unit} \times \text{m}^3 \text{s}^{-1}]$.

The negative directional row is already signed. Therefore net transport equals positive plus negative.

Step 1 — Read the native water net transport

For the River/Main interface:

$$Q_{0/1,\text{raw}}(t) \quad [\text{m}^3 \text{s}^{-1}]$$

For the Ocean/Main interface:

$$Q_{1/2,\text{raw}}(t) \quad [\text{m}^3 \text{s}^{-1}]$$

Step 2 — Re-orient the River sign

$$Q_{\text{River,in}}(t) = -Q_{0/1,\text{raw}}(t)$$

Step 3 — Retain the Ocean sign

$$Q_{\text{Ocean,in}}(t) = +Q_{1/2,\text{raw}}(t)$$

Step 4 — Calculate the net flux into the Main CV

$$Q_{\text{net}}(t) = Q_{\text{River,in}}(t) + Q_{\text{Ocean,in}}(t) \quad [\text{m}^3 \text{s}^{-1}]$$

Step 5 — Integrate over time

Because the flux.out time column is in days while Q is in $\text{m}^3 \text{s}^{-1}$, each time increment is converted to seconds before integration.

$$\Delta V_{\text{RHS,C}} = \int Q_{\text{net}}(t) dt \quad [\text{m}^3]$$

All Region-0/1 interface edges are accumulated into one native River/Main transport. Method C is therefore independent of the explicit edge-by-edge Method-B reconstruction.

2.5 Water closure checks

Three checks are available for the Main CV:

1. Storage consistency

$$\Delta V_{\text{LHS,1}} \approx \Delta V_{\text{LHS,2}}$$

2. Flux consistency

$$\Delta V_{\text{RHS,B}} \approx \Delta V_{\text{RHS,C}}$$

3. Water-budget closure

$$\Delta V_{\text{LHS}} \approx \Delta V_{\text{RHS}}$$

3. Salt budget

Salt uses the same control-volume geometry and boundary pathways as Water. The difference is that the transported or stored water volume is weighted by salinity. The final budget unit is **tonnes equivalent salt**.

3.1 Salt unit conversion

The adopted equivalent-salt conversion used by the budget is:

$$1 \text{ PSU m}^3 \approx 0.001025 \text{ tonnes equivalent salt}$$

The conversion can be understood from the adopted water density $\rho_w \approx 1025 \text{ kg m}^{-3}$ and the budget approximation $1 \text{ PSU} \approx 1 \text{ g kg}^{-1}$.

Step 1 — Convert one PSU in one cubic metre to grams

$$1 \text{ g kg}^{-1} \times 1025 \text{ kg} = 1025 \text{ g}$$

Step 2 — Convert grams to kilograms

$$1025 \text{ g} = 1.025 \text{ kg}$$

Step 3 — Convert kilograms to tonnes

$$1.025 \text{ kg} = 0.001025 \text{ tonnes}$$

Define the conversion factor:

$$f_s = 0.001025 \frac{\text{tonnes}}{\text{PSU m}^3}$$

This conversion is applied only after salinity has been multiplied by an associated water volume. PSU alone is not converted directly to mass.

3.2 LHS-1: node-based salt inventory

LHS-1 integrates the saved nodal salinity over the nodal water column and then over each horizontal triangle.

Source files

- salinity_*.nc: nodal salinity $S_{i,k}(t)$ [PSU].
- zCoordinates_*.nc: nodal vertical coordinates $z_{i,k}(t)$ [m].
- hgrid.gr3: element geometry and area A_e [m²].
- out2d_*.nc: dryFlagElement.

Step 1 — Calculate the thickness of one nodal vertical interval

$$h_{i,k} = |z_{i,k+1} - z_{i,k}| \quad [\text{m}]$$

Step 2 — Calculate mean salinity in that interval

$$\bar{S}_{i,k} = \frac{S_{i,k} + S_{i,k+1}}{2} \quad [\text{PSU}]$$

Step 3 — Calculate salinity content per unit horizontal area for that interval

$$I_{S,i,k} = \bar{S}_{i,k} h_{i,k} \quad [\text{PSU m}]$$

Step 4 — Sum the whole water column at one node

$$I_{S,i} = \sum_k I_{S,i,k} \quad [\text{PSU m}]$$

Step 5 — Convert the three nodal columns to one triangular-element salt content

For triangular element e with nodes 1, 2 and 3:

$$C_{S,e} = (1 - D_e) A_e \frac{I_{S,1} + I_{S,2} + I_{S,3}}{3} \quad [\text{PSU m}^3]$$

Step 6 — Convert the element content to tonnes equivalent salt

$$M_{S,e} = f_s C_{S,e} \quad [\text{tonnes}]$$

Step 7 — Sum all elements

$$M_{S,\text{LHS1}}(t) = \sum_e M_{S,e}(t) \quad [\text{tonnes}]$$

Step 8 — Calculate the LHS-1 storage change

$$\Delta M_{S,LHS1} = M_{S,LHS1}(t_{\text{end}}) - M_{S,LHS1}(t_{\text{begin}}) \quad [\text{tonnes}]$$

3.3 RHS Method B: reconstructed salt transport

Method B uses the edge-layer water flux $q_{s,k}$ already defined in Water Method B and multiplies it by representative salinity for the same edge layer.

Step 1 — Calculate mean salinity over the vertical interval at node i

$$\bar{S}_{i,k} = \frac{S_{i,k} + S_{i,k+1}}{2} \quad [\text{PSU}]$$

Step 2 — Do the same at node j

$$\bar{S}_{j,k} = \frac{S_{j,k} + S_{j,k+1}}{2} \quad [\text{PSU}]$$

Step 3 — Average salinity to the edge layer

$$\bar{S}_{s,k} = \frac{\bar{S}_{i,k} + \bar{S}_{j,k}}{2} \quad [\text{PSU}]$$

Step 4 — Multiply edge-layer salinity by edge-layer water flux

$$F_{S,s,k} = \bar{S}_{s,k} q_{s,k} \quad [\text{PSU m}^3 \text{ s}^{-1}]$$

Unit check:

$$[\text{PSU}] \times [\text{m}^3 \text{ s}^{-1}] = [\text{PSU m}^3 \text{ s}^{-1}]$$

Step 5 — Sum all layers and all selected boundary edges

$$F_{S,\text{net}}(t) = \sum_s \sum_k F_{S,s,k}(t) \quad [\text{PSU m}^3 \text{ s}^{-1}]$$

Step 6 — Integrate over time

$$I_{S,\text{bnd},B} = \int F_{S,\text{net}}(t) dt \quad [\text{PSU m}^3]$$

Step 7 — Convert to tonnes equivalent salt

$$M_{S,\text{bnd},B} = f_S I_{S,\text{bnd},B} \quad [\text{tonnes}]$$

For the current Main-CV accounting, rainfall, evaporation and direct Region-1 salt source/sink terms are zero.

3.4 RHS Method C: native SCHISM salinity transport

Method C reads native salinity transport from outputs/flux.out. Rows 5 and 6 of each iflux = 2 block contain positive- and negative-direction salinity transport.

Step 1 — Form native net salinity transport for each interface

The negative row is already signed:

$$F_{S,\text{raw}} = F_{S,\text{pos}} + F_{S,\text{neg}} \quad [\text{PSU m}^3 \text{ s}^{-1}]$$

Step 2 — Convert the River interface to positive-into-Main-CV

$$F_{S,\text{River},\text{in}} = -F_{S,0/1,\text{raw}}$$

Step 3 — Retain the Ocean interface sign

$$F_{S,\text{Ocean},\text{in}} = +F_{S,1/2,\text{raw}}$$

Step 4 — Calculate net salinity transport into the Main CV

$$F_{S,\text{net}} = F_{S,\text{River},\text{in}} + F_{S,\text{Ocean},\text{in}} \quad [\text{PSU m}^3 \text{ s}^{-1}]$$

Step 5 — Integrate using time in seconds

$$I_{S,\text{bnd},C} = \int F_{S,\text{net}}(t) dt \quad [\text{PSU m}^3]$$

Step 6 — Convert to tonnes equivalent salt

$$M_{S,\text{bnd},C} = f_S I_{S,\text{bnd},C} \quad [\text{tonnes}]$$

3.5 Salt closure checks

Two checks are retained for the Main CV:

1. Flux consistency

$$M_{S,\text{bnd},B} \approx M_{S,\text{bnd},C}$$

2. Salt-budget closure

For either RHS method:

$$R_S = \Delta M_{S,\text{LHS}} - \Delta M_{S,\text{RHS}} \quad [\text{tonnes}]$$

$$R_S(\%) = 100 \frac{R_S}{|\Delta M_{S,\text{RHS}}|}$$

4. Total Nitrogen (TN) budget

TN is reactive. Its budget contains boundary transport together with physical and biogeochemical sources and sinks inside the selected control volume. The final budget unit is **tonnes of nitrogen**, written as t N.

The accounting is kept in the familiar form:

$$\text{Storage change} = \text{IN} - \text{OUT} + \text{Sources} - \text{Sinks}$$

All individual terms are converted to one common sign convention before they are added: positive increases water-column TN inside the control volume; negative decreases it.

4.1 TN definition and unit conversion

The TN concentration used in the accounting is:

$$C_{\text{TN}} = \text{NIT_amm} + \text{NIT_nit} + \text{OGM_don} + \text{OGM_pon} + 0.15 \text{ PHY_mixed}$$

After the PHY conversion:

$$C_{\text{TN}} \quad [\text{mmol N m}^{-3}]$$

The factor used to convert mmol N to tonnes N is derived step by step.

Step 1 — Convert mmol N to mol N

$$1 \text{ mmol N} = 10^{-3} \text{ mol N}$$

Step 2 — Use the molar mass of nitrogen

$$1 \text{ mol N} = 14.0067 \text{ g N}$$

Therefore:

$$1 \text{ mmol N} = 0.0140067 \text{ g N}$$

Step 3 — Convert grams to tonnes

$$1 \text{ tonne} = 10^6 \text{ g}$$

Hence:

$$1 \text{ mmol N} = 14.0067 \times 10^{-9} \text{ t N}$$

Define:

$$f_N = 14.0067 \times 10^{-9} \frac{\text{t N}}{\text{mmol N}}$$

4.2 LHS-1: node-based TN inventory

LHS-1 reconstructs water-column TN from the nodal TN constituent fields and nodal vertical coordinates.

Source files

- TN tracer outputs containing NIT_amm, NIT_nit, OGM_don, OGM_pon, and PHY_mixed: nodal constituent concentrations.
- zCoordinates_*.nc: nodal vertical coordinates [m].
- hgrid.gr3: element area A_e [m²].
- out2d_*.nc: dryFlagElement.

Step 1 — Construct TN concentration at each node and vertical level

At node i and level k :

$$C_{\text{TN},i,k} = \text{NIT_amm}_{i,k} + \text{NIT_nit}_{i,k} + \text{OGM_don}_{i,k} + \text{OGM_pon}_{i,k} + 0.15 \text{ PHY_mixed}_{i,k}$$

$$C_{\text{TN},i,k} \quad [\text{mmol N m}^{-3}]$$

Step 2 — Calculate the thickness of one nodal vertical interval

$$h_{i,k} = |z_{i,k+1} - z_{i,k}| \quad [\text{m}]$$

Step 3 — Calculate mean TN concentration in that interval

$$\bar{C}_{TN,i,k} = \frac{C_{TN,i,k} + C_{TN,i,k+1}}{2} \quad [\text{mmol N m}^{-3}]$$

Step 4 — Calculate TN amount per unit horizontal area for that interval

$$I_{TN,i,k} = \bar{C}_{TN,i,k} h_{i,k} \quad [\text{mmol N m}^{-2}]$$

Step 5 — Sum the whole water column at one node

$$I_{TN,i} = \sum_k I_{TN,i,k} \quad [\text{mmol N m}^{-2}]$$

Step 6 — Convert the three nodal columns to one element TN amount

For triangular element e with nodes 1, 2 and 3:

$$N_{TN,e} = (1 - D_e) A_e \frac{I_{TN,1} + I_{TN,2} + I_{TN,3}}{3} \quad [\text{mmol N}]$$

Unit check:

$$[\text{mmol N m}^{-2}] \times [\text{m}^2] = [\text{mmol N}]$$

Step 7 — Convert the element TN amount to tonnes N

$$M_{TN,e} = f_N N_{TN,e} \quad [\text{t N}]$$

Step 8 — Sum all elements

$$M_{TN,LHS1}(t) = \sum_e M_{TN,e}(t) \quad [\text{t N}]$$

Step 9 — Calculate the LHS-1 TN storage change

$$\Delta M_{TN,LHS1} = M_{TN,LHS1}(t_{\text{end}}) - M_{TN,LHS1}(t_{\text{begin}}) \quad [\text{t N}]$$

4.3 LHS-2: native AED element-layer TN inventory

LHS-2 uses the native AED element-layer TN concentration and AED layer thickness. No nodal thickness reconstruction is required.

Source files

- aed_data_cmb_*.nc: TOT_tn(t,e,k) [mmol N m⁻³] and ENV_layer_ht(t,e,k) [m].
- hgrid.gr3: element area A_e [m²].

Step 1 — Read TN concentration in one AED element layer

$$C_{TN,e,k} = \text{TOT_tn}(e, k) \quad [\text{mmol N m}^{-3}]$$

Step 2 — Read the AED layer thickness

$$h_{e,k} = \text{ENV_layer_ht}(e, k) \quad [\text{m}]$$

Step 3 — Calculate the volume of one element-layer prism

$$V_{e,k} = A_e h_{e,k} \quad [\text{m}^3]$$

Step 4 — Calculate the amount of nitrogen in that prism

$$N_{e,k} = C_{TN,e,k} V_{e,k} \quad [\text{mmol N}]$$

Unit check:

$$[\text{mmol N m}^{-3}] \times [\text{m}^3] = [\text{mmol N}]$$

Step 5 — Convert the prism amount to tonnes N

$$M_{TN,e,k} = f_N N_{e,k} \quad [\text{t N}]$$

Step 6 — Sum all layers and elements

$$M_{TN,LHS2}(t) = \sum_e \sum_k M_{TN,e,k}(t) \quad [\text{t N}]$$

Step 7 — Calculate the LHS-2 TN storage change

$$\Delta M_{TN,LHS2} = M_{TN,LHS2}(t_{\text{end}}) - M_{TN,LHS2}(t_{\text{begin}}) \quad [\text{t N}]$$

TN storage consistency check

$$\Delta M_{TN,LHS1} \approx \Delta M_{TN,LHS2}$$

4.4 RHS Method B: reconstructed TN boundary transport

Method B uses the same edge-layer water flux $q_{s,k}$ defined in Water Method B. TN concentration is assembled at the same nodes and levels and then averaged to the same edge layer.

Source files

- horizontalVelX_*.nc, horizontalVelY_*.nc: nodal velocity.
- zCoordinates_*.nc: nodal vertical coordinates.
- TN tracer outputs: NIT_amm, NIT_nit, OGM_don, OGM_pon, PHY_mixed.
- hgrid.gr3 and fluxflag.prop: edge geometry and Main-CV interface identification.

Step 1 — Construct nodal TN concentration

At node i and level k :

$$C_{TN,i,k} = \text{NIT_amm} + \text{NIT_nit} + \text{OGM_don} + \text{OGM_pon} + 0.15 \text{ PHY_mixed}$$

$$C_{TN,i,k} \quad [\text{mmol N m}^{-3}]$$

The same calculation is performed at node j .

Step 2 — Average TN concentration over the vertical interval at node i

$$\bar{C}_{TN,i,k} = \frac{C_{TN,i,k} + C_{TN,i,k+1}}{2} \quad [\text{mmol N m}^{-3}]$$

Step 3 — Average TN concentration over the vertical interval at node j

$$\bar{C}_{TN,j,k} = \frac{C_{TN,j,k} + C_{TN,j,k+1}}{2} \quad [\text{mmol N m}^{-3}]$$

Step 4 — Average the two nodal values to the edge layer

$$\bar{C}_{TN,s,k} = \frac{\bar{C}_{TN,i,k} + \bar{C}_{TN,j,k}}{2} \quad [\text{mmol N m}^{-3}]$$

Step 5 — Multiply by the edge-layer water flux

$$F_{TN,s,k} = \bar{C}_{TN,s,k} q_{s,k} \quad [\text{mmol N s}^{-1}]$$

Unit check:

$$[\text{mmol N m}^{-3}] \times [\text{m}^3 \text{ s}^{-1}] = [\text{mmol N s}^{-1}]$$

Step 6 — Sum all layers and boundary edges

$$F_{TN,\text{net}}(t) = \sum_s \sum_k F_{TN,s,k}(t) \quad [\text{mmol N s}^{-1}]$$

Step 7 — Integrate over time in seconds

$$N_{TN,\text{bnd},B} = \int F_{TN,\text{net}}(t) dt_s \quad [\text{mmol N}]$$

Step 8 — Convert boundary transport to tonnes N

$$M_{TN,\text{bnd},B} = f_N N_{TN,\text{bnd},B} \quad [\text{t N}]$$

Step 9 — Complete the RHS accounting

Start with the cumulative boundary transport from Method B. Then add the signed Main-CV terms one by one:

1. direct Region-1 point source / sink: 0 for the present setup;
2. gaseous nitrogen loss: sink, therefore negative;
3. atmospheric DIN: retain the saved sign;
4. sediment-water exchange: retain the saved sign.

The final RHS therefore follows the same physical balance stated above:

$$\Delta M_{TN,\text{RHS},B} = \text{IN} - \text{OUT} + \text{Sources} - \text{Sinks} \quad [\text{t N}]$$

4.5 RHS Method C: native SCHISM TN tracer transport

Method C reads the native positive/negative tracer-transport pairs from outputs/flux.out and constructs TN transport from the same constituent pools used in the TN definition.

Step 1 — Form native net transport for each tracer X

The negative directional row is already signed:

$$F_{X,\text{raw}} = F_{X,\text{pos}} + F_{X,\text{neg}}$$

Step 2 — Assemble native TN transport for one interface

$$F_{TN,raw} = F_{NIT_amm} + F_{NIT_nit} + F_{OGM_don} + F_{OGM_pon} + 0.15F_{PHY_mixed}$$

$$F_{TN,raw} \quad [mmol\ N\ s^{-1}]$$

Step 3 — Re-orient the River interface

$$F_{TN,River,in} = -F_{TN,0/1,raw}$$

Step 4 — Retain the Ocean interface sign

$$F_{TN,Ocean,in} = +F_{TN,1/2,raw}$$

Step 5 — Calculate the net native TN transport into the Main CV

$$F_{TN,net} = F_{TN,River,in} + F_{TN,Ocean,in} \quad [mmol\ N\ s^{-1}]$$

Step 6 — Integrate in seconds

$$N_{TN,bnd,C} = \int F_{TN,net}(t) dt_s \quad [mmol\ N]$$

Step 7 — Convert to tonnes N

$$M_{TN,bnd,C} = f_N N_{TN,bnd,C} \quad [t\ N]$$

Step 8 — Complete the RHS accounting

Start with the cumulative native boundary transport from Method C. Add the same signed Main-CV source and sink terms used with Method B.

$$\Delta M_{TN,RHS,C} = IN - OUT + Sources - Sinks \quad [t\ N]$$

Thus Method B and Method C differ only in how the boundary transport is obtained; the internal source and sink accounting is the same.

4.6 TN closure checks**1. Storage consistency**

$$\Delta M_{TN,LHS1} \approx \Delta M_{TN,LHS2}$$

2. Flux consistency

$$M_{TN,bnd,B} \approx M_{TN,bnd,C}$$

3. TN-budget closure

For either RHS transport method:

$$R_{TN} = \Delta M_{TN,LHS} - \Delta M_{TN,RHS} \quad [t\ N]$$

$$R_{TN}(\%) = 100 \frac{R_{TN}}{|\Delta M_{TN,RHS}|}$$

5. Calculation summary**Water**

- LHS-1: SCHISM node-based water storage.
- LHS-2: AED element-layer water storage.
- Method B: reconstructed edge-layer water flux.
- Method C: native VOL transport from flux.out.
- Final unit: m³.

Salt

- LHS-1: nodal salinity integrated over the nodal water column and horizontal elements.
- Method B: edge-layer salinity multiplied by reconstructed water flux.
- Method C: native salinity transport from flux.out.
- Final unit: tonnes equivalent salt.

TN

- LHS-1: nodal TN constituent fields integrated with zCoordinates_*.nc and element area.
- LHS-2: native TOT_tn multiplied by AED element-layer volume.
- Method B: reconstructed TN boundary transport plus the signed Main-CV source and sink terms.
- Method C: native TN-pool boundary transport plus the same signed Main-CV source and sink terms.

- Final unit: t N.

The two RHS pathways are **alternative independent transport calculations and are not additive**. Method B reconstructs transport from saved model states and geometry. Method C uses SCHISM runtime region-interface transport diagnostics.

For every closure comparison, the LHS and RHS must use the **same spatial control volume**, the **same begin/end times**, the **same sign convention**, and compatible **final units**.

Appendix A. Source files and primary variables

The key source files are:

- hgrid.gr3: node coordinates, bathymetry and element connectivity. Used for element area, edge length, normals and geometry [m, m²].
- out2d_*.nc: elevation and dryFlagElement. Used for Water LHS-1 and wet/dry masks [m, -].
- aed_data_cmb_*.nc: ENV_layer_ht, TOT_tn and TN source/sink diagnostics. Used for Water LHS-2, TN LHS-2 and TN source/sink terms [m; mmol N m⁻³; source/sink rate units].
- horizontalVelX_*.nc: nodal *u* velocity for Method B [m s⁻¹].
- horizontalVelY_*.nc: nodal *v* velocity for Method B [m s⁻¹].
- zCoordinates_*.nc: nodal vertical coordinates for Method B, Salt LHS-1 and TN LHS-1 [m].
- salinity_*.nc: nodal salinity for Salt LHS-1 and Salt Method B [PSU].
- TN tracer outputs: NIT_amm, NIT_nit, OGM_don, OGM_pon, PHY_mixed. Used for TN LHS-1 and TN Method B; assembled to mmol N m⁻³.
- fluxflag.prop: element region IDs for Main-CV native interface definition.
- outputs/flux.out: native water, temperature, salinity and tracer transport for Method C [quantity-specific transport units].

Appendix B. Unit-conversion checklist

Before accepting any budget result, verify the following unit chain explicitly.

Water

$$\text{m}^3 \text{ s}^{-1} \times \text{s} = \text{m}^3$$

Salt

$$\begin{aligned} \text{PSU} \times \text{m}^3 &= \text{PSU m}^3 \\ \text{PSU m}^3 \times 0.001025 \frac{\text{tonnes}}{\text{PSU m}^3} &= \text{tonnes} \end{aligned}$$

TN boundary transport

$$\begin{aligned} \text{mmol N m}^{-3} \times \text{m}^3 \text{ s}^{-1} &= \text{mmol N s}^{-1} \\ \text{mmol N s}^{-1} \times \text{s} &= \text{mmol N} \\ \text{mmol N} \times 14.0067 \times 10^{-9} \frac{\text{t N}}{\text{mmol N}} &= \text{t N} \end{aligned}$$

TN volumetric sink/source rate

$$\text{mmol N m}^{-3} \text{ d}^{-1} \times \text{m}^3 = \text{mmol N d}^{-1}$$

TN sheet-flux source/sink rate

$$\text{mmol N m}^{-2} \text{ d}^{-1} \times \text{m}^2 = \text{mmol N d}^{-1}$$

For both TN source/sink rate forms, multiply by f_N to obtain t N d⁻¹ and integrate using time in days to obtain t N.